

## **Positive Phototropism**

Positive phototropism is typical of aerial organs, especially stems. Growth occurs in the direction of light, allowing the plant to maximize light capture for photosynthesis.

## **Negative Phototropism**

Some roots exhibit this response by growing away from light. This adaptive behavior allows roots to grow into the soil, where they can absorb water and nutrients.

---

## **3. REQUIRED MATERIALS**

### **List of materials**

For a class of 30 students divided into groups of 3–4:

#### **Materials for each group:**

- 2 medium-sized cardboard boxes (approximately 30×20×20 cm)
- Scissors or cutter (under supervision)
- Black adhesive tape
- Aluminum foil
- Graph paper
- Ruler or measuring tape
- Pencils and markers
- Camera or smartphone for documentation
- Desk lamp or flashlight
- Small containers for seeds (yogurt cups or plastic cups)
- Potting soil
- Fast-germinating seeds (beans, lentils, cress)
- Spray bottle for water
- Disposable gloves

#### **Common materials for the class:**

- Precision scale
- Room thermometer
- Lux meter (if available)

- Computer access for data processing
  - Microscope (optional, for cellular observations)
- 

### **Low-cost alternatives**

If the budget is limited, the following alternatives can be used:

- Use shoeboxes instead of larger boxes
  - Use recycled yogurt containers or plastic cups for planting
  - Use common, inexpensive seeds such as beans, which have a high germination rate
  - Replace the lux meter with free smartphone apps that measure light intensity
  - Use mobile phones for photographic documentation
- 

### **Preliminary preparation**

Before starting the experiment in class:

1. Check the germination rate of the seeds with a preliminary test
  2. Prepare the cardboard boxes by cutting the necessary openings
  3. Gather all materials and organize them for each group
  4. Set up a dedicated area in the classroom where the plants can grow undisturbed
  5. Prepare worksheets for data collection
- 

## **4. DETAILED PROCEDURE**

### **Preparation of the boxes**

#### **Box 1: Control (normal growth)**

1. Take a cardboard box and completely remove the top lid
2. Cover the inside with aluminum foil to maximize light reflection
3. Place the box in an area with uniform lighting

#### **Box 2: Phototropism experiment**

1. Take a second cardboard box of the same size
2. Seal all openings with black tape to prevent light from entering
3. Create an opening of about 5×5 cm on one side (preferably at the top)

4. Reinforce the edges of the opening with tape
  5. Cover the inside of the box with aluminum foil
  6. Clearly label the boxes as “Control” and “Phototropism experiment”
- 

### **Sowing and germination**

1. Fill the small containers with soil up to about 3/4 of their capacity
  2. Moisten the soil with water (it should be damp but not soaked)
  3. Place 3–4 seeds in each container, about 1 cm deep
  4. Lightly cover the seeds with more soil
  5. Label each container with the group name and sowing date
  6. Place all containers in a well-lit area, but not in direct sunlight
  7. Keep the soil consistently moist by spraying water when necessary
  8. Wait for seed germination (about 3–7 days, depending on the species)
- 

### **Experimental setup**

#### **Phase 1: Initial arrangement**

1. When sprouts reach a height of 2–3 cm, select the most uniform seedlings
2. Measure and record the initial height and orientation of each seedling
3. Take photos from above and from the side to document the initial condition

#### **Phase 2: Placement in the boxes**

1. Place half of the containers in the control box (uniform light)
2. Place the other half in the phototropism experiment box
3. In the experimental box, ensure that light enters only through the side opening
4. Position the lamp so that it illuminates the opening of the experimental box

#### **Phase 3: Environmental conditions**

1. Ensure both boxes are exposed to the same temperature
  2. Measure and record light intensity in both boxes
  3. Establish an identical watering schedule for both groups
-

## **Monitoring and data collection**

1. Observe the seedlings daily for a period of 10–14 days
  2. Measure and record the following parameters:
    - Seedling height (from soil to apex)
    - Curvature angle relative to the vertical (only for the experimental group)
    - Number and size of leaves
    - Color of leaves and stem
  3. Take daily photographs from the same angle and distance
  4. Record any observable changes or anomalies
- 

## **5. OBSERVATION PROTOCOL**

### **Recommended timing**

- Daily observations: take measurements at the same time each day
- Experiment duration: 2 weeks (10–14 days)
- Intensive observation sessions: at the beginning, day 3, day 7, and at the end

To optimize classroom time, an observation schedule can be organized so that different student groups take turns performing daily measurements.

---

### **Parameters to measure**

#### **Quantitative parameters:**

- Seedling height (mm)
- Stem diameter (mm)
- Curvature angle relative to vertical (degrees)
- Number of leaves
- Leaf surface area (mm<sup>2</sup>)
- Distance between nodes (mm)
- Light intensity (lux)
- Ambient temperature (°C)

#### **Qualitative parameters:**

- Stem color
  - Leaf color
  - Overall plant vigor
  - Presence of anomalies or diseases
- 

### **Photographic documentation**

1. Create a fixed station for photos, with a neutral background and constant lighting
2. Use a ruler or reference scale in each photo
3. Take:
  - Top view photos
  - Side view photos (90° relative to growth direction)
  - Close-ups of leaves and relevant parts
4. Organize photos in a temporal sequence to create a time-lapse of growth